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Printed Pages – 8

324811 (24)

BE (8th Semester)

Examination, April - May, 2013

Branch : Electrical

MODERN CONTROL SYSTEM

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Answer all questions. Part (a) of each question is compulsory. Do any two from (b), (c) and (d) of each question. Do (b) or (c) of question V.

Q. 1. a) What are stable and unstable limit cycles ?

b) What are describing functions ? Derive the describing function of a relay with dead zone and hysteresis.

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(c) Consider a third order system with a saturating amplifier of the Figure 1 having gain k in its linear region. Determine the largest value of gain K for the system to stay stable. What would be frequency, amplitude and nature of the limit cycle for a gain of $k = 3$.

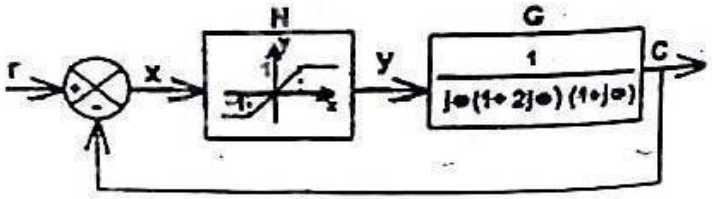

Figure 1

(d) State and prove Liapunov's theorem of stability for linear systems. For the spring, mass damper system shown in Figure 2

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determine the equilibrium state and show that it is stable.

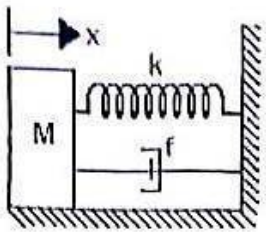

Figure 2

Q. 2. (a) Consider a state model $\dot{x} = Ax + Bu$, where

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -40 & -34 & -10 \end{bmatrix}; \; B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Find the eigen values of A.

(b) A system has the following transfer function

$$\frac{Y(s)}{U(s)} = \frac{2s^2 + 6s + 7}{(s+1)^2(s+2)}$$

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Obtain the Jordon Canonical form representation. Draw the state diagram of the above and determine the controllability of the system.

(c) Investigate the observability of the system

$$\dot{x} = \begin{bmatrix} 1 & e^{-t} \\ 0 & 1 \end{bmatrix} x \quad y = [1 \;\; -1] x$$

Is the above system controllable and observable at t = 0 ? If yes find x(0) when $y(t) = e^t$. Find x(t).

(d) Prove that for

$$A = \begin{bmatrix} -a_1 & 1 & 0 & \cdots & 0 & 0 \\ -a_2 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ -a_{n-1} & 0 & 0 & \cdots & 0 & 1 \\ -a_n & 0 & 0 & \cdots & 0 & 0 \end{bmatrix}$$

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with distinct eigen values $\lambda_1, \lambda_2, \ldots, \lambda_n$, the diagonalizing matrix is given by

$$M = \begin{bmatrix} 1 & \cdots & 1 \\ \lambda_1 + a_1 & \cdots & \lambda_n + a_1 \\ \lambda_1^2 + a_1\lambda_1 + a_2 & \cdots & \lambda_n^2 + a_1\lambda_n + a_2 \\ \vdots & & \vdots \\ \lambda_1^{n-1} + a_1\lambda_1^{n-2} + \cdots + a_{n-2}\lambda_1 + a_{n-1} & \cdots & \lambda_n^{n-1} + a_1\lambda_n^{n-2} + \cdots + a_{n-2}\lambda_n + a_{n-1} \end{bmatrix}$$

Q. 3. (a) Write the necessary and sufficient condition for arbitrary pole placement.

(b) Consider the state equation $\dot{x} = Ax + Bu$, where

$$A = \begin{bmatrix} 1 & 1 \\ -4 & -3 \end{bmatrix} \qquad B = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$$

The following design specifications are to be met $\zeta = 0.5$, $w_n = 2$ rad/sec. Determine the necessary feedback gain matrix K and the control signal u.

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(c) Derive Ackermann's formula for the determination of the state feedback gain matrix K and state observer gain matrix K_e.

(d) Explain the terms full order and minimum order state observers. For a state model with

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad C = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

Assuming that the output y can be accurately measured, design a reduced order observer with the desired eigen values for the reduced order observer as

$\mu_1 = -2 + j3.464$ and $\mu_2 = -2 - j3.464$

Q. 4. (a) State Shanon's sampling theorem. What is aliasing of signal ?

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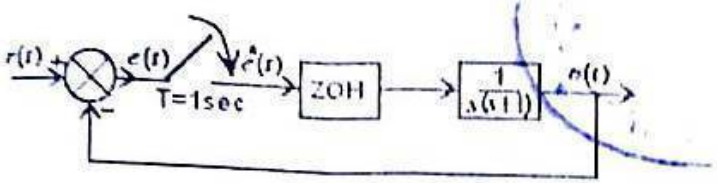

Figure 3

(b) What is a zero-order hold (ZOH)? What is the transfer function of ZOH? For the sampled data control system shown in figure 3, find the output C(z) when the input is a unit step input taking the sampling period T = 1sec.

(c) Determine the stability of the following characteristic equations using Jury's test

(i) $Q(z) = z^3 + z^2 + 0.2z + 0.5$

(ii) $Q(z) = z^3 + z^2 + 0.2z - 0.5$

(d) Determine the range of gain K, for stability for a system with the following characteristic equation

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$$z^2 + (0.55K - 1.55)z + (0.55 + 0.25K) = 0$$

Find the frequency of steady state oscillations.

Q. 5. (a) State the optimal control problem for discrete time systems using the transfer function and the state variable approaches.

(b) Derive the discrete Ricatti equation.

(c) The linear discrete system

$$x_1(k+1) = x_1(k) + x_2(k)$$

$$x_2(k+1) = x_2(k) + u(k)$$

is to be controlled to minimize the performance index

$$J = \sum_{k=0}^{2} [4x_1^2(k) + u^2(k)]$$

Obtain the control sequence [u(0), u(1), u(2)] with the initial state as $x(0) = [1 \quad 0]^T$.

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